Fire-Aware Robot Dogs

Machine-Learning Wildfire Forecasts for Safe Navigation and Scouting

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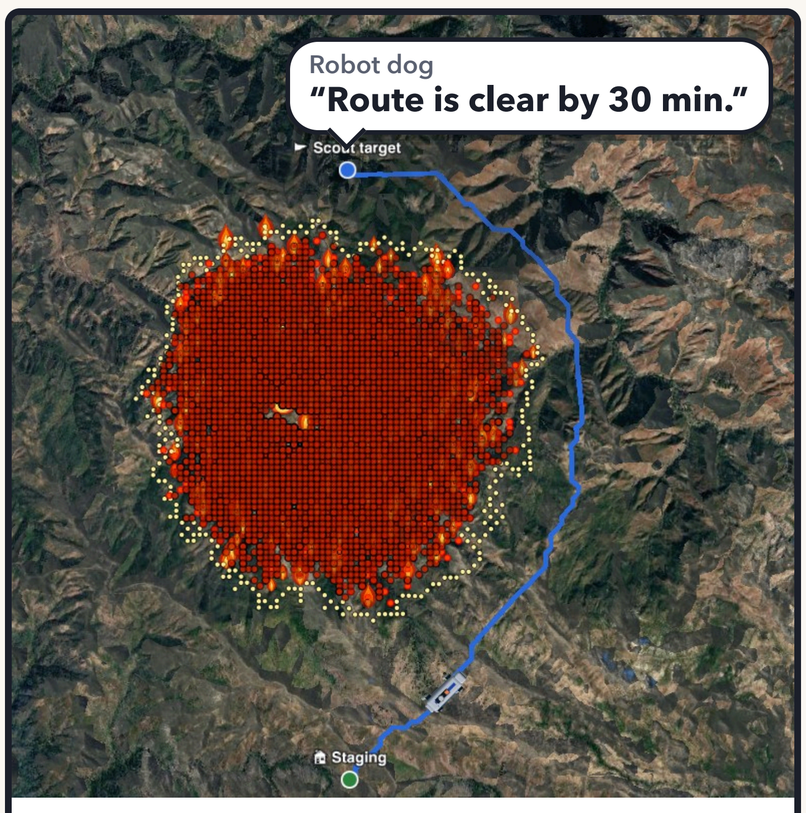
One afternoon at a Big Sur fire

A simulated mission. Every picture is from our simulator.



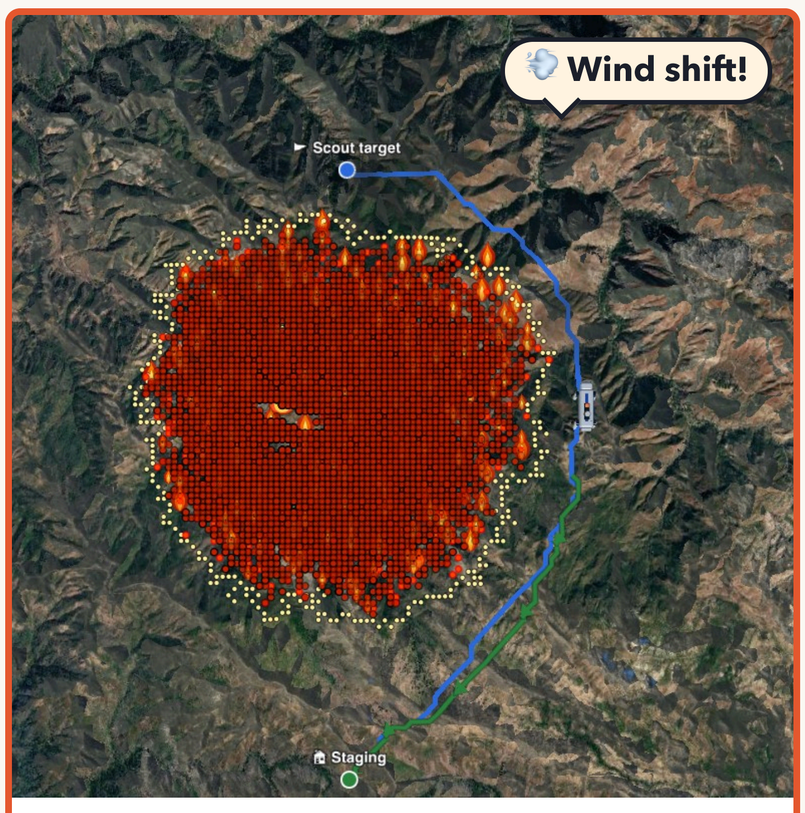
3:00 PM

The fire has burned since midnight on steep, brushy slopes. The crew needs to know **where the edge is going**.



3:00 PM • Plan

On its forecast, the dog plans a route (**blue**) that passes every spot **30+ min before the fire**, slower through brush and uphill.



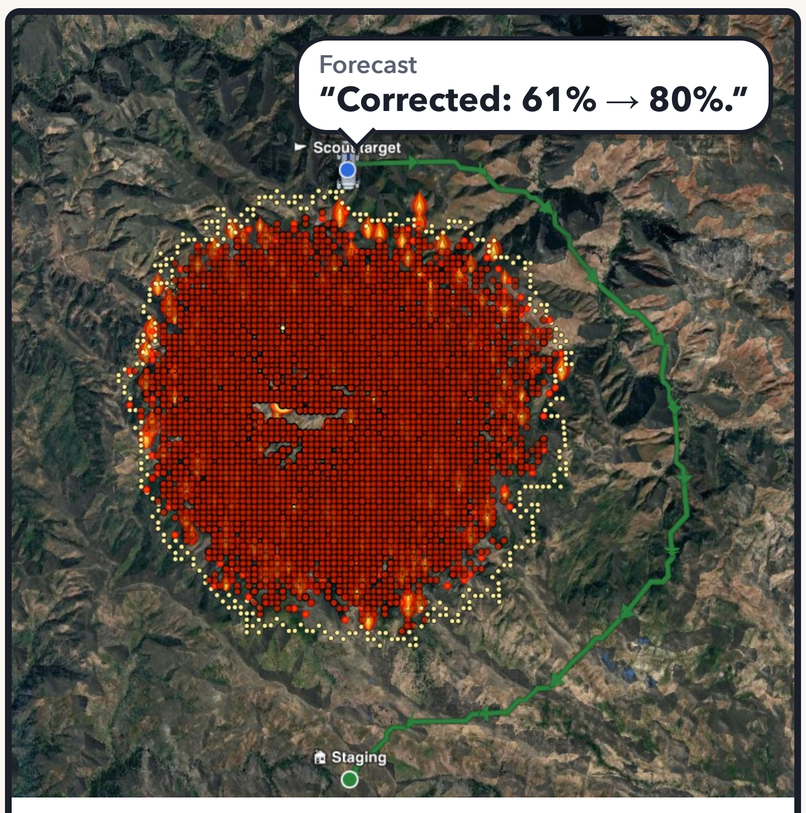
3:30 PM • Surprise

The wind shifts and the real fire turns. **The forecast doesn't know**: match 82% → 61%. The escape path home (**green**) stays open.



3:45 PM • Measure

Its sensors see the real fire coming. It measures **wind, temperature, humidity and the fire edge** and sends them back.



4:45 PM • Scout

On the **corrected forecast** the dog re-plans, reaches the fire edge and scouts it.

✓ Home safe

0 people sent to the fire edge

Forecast match **61% → 80%**

Fire margin kept: **≥ 30 min**

~8 PM • Home

Back at staging before the fire arrives. The crew plans with a **better forecast**.

The loop behind the story: **Predict** → **Safe route** → **Escape path** → **Scout** → **Feedback** ↺

Know where the fire *will* be – not just where it is.

- **Safer crews** — Fewer firefighters sent into danger to look.
- **Robot stays safe** — Routes and exits keep it from getting trapped.
- **Smarter sensing** — Measure where it changes decisions.